Cross Sensor Transfer Learning for Unsupervised SAR Target Detection

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Abstract—**Synthetic Aperture Radar (SAR) target detection based on deep learning typically assume the training and test data drawn from the same distribution. However, when the radar sensors change, this assumption does not always hold and will cause to a performance degradation when encountering such distribution mismatch. Transfer learning shares a great idea to solve this issue by transferring knowledge from the labeled data of source sensor to the unlabeled data of target sensor. We proposed a cross sensor transfer learning method based on domain adaptation, which can improve the cross sensor robustness of SAR target detection with unlabeled target sensor (domain) data. Our method includes three stages: pixel domain diverse (PDD), multilevel feature alignment (MFA) and iterative self-training (IST). The PDD stage can alleviate the few training data of SAR target detection and enhance the generalization ability by generating transition domain. MFA can learn domain invariant features in a multi-layer feature space by using the idea of adversarial learning. To learn more discriminative features, we further proposed the IST that can generate and select high-quality pseudo labels for the SAR images of target sensor. The experimental results on miniSAR and FARADSAR datasets demonstrate the effectiveness of the proposed approach.**

Keywords—adversarial learning, iterative self-training Synthetic Aperture Radar (SAR), cross sensor transfer learning, unsupervised target detection, domain adaptation.

I. Introduction

In recent years, deep learning based on convolutional neural networks (CNNs) has achieved great success in Synthetic Aperture Radar (SAR) target detection. And a large amount of target detection methods based on CNNs have been widely used in SAR image target detection. Due to its flexibility and scalability, Faster R-CNN [1] has become one of the most popular methods.

Though these SAR target detection methods based on deep learning methods have achieved great performance, they typically assume the training and test data drawn from the same distribution. It is difficult to hold when the radar sensors change with the variance in the parameters, viewpoints, scenes, etc. When the training samples and test samples are obtained by the same radar for the same task in practice, this assumption holds. But if we first annotate the data (training data) acquired by one radar, and then obtain the unlabeled data (test data) for the same task of another radar, the distribution between them are different, which will cause a obviously performance degradation. Annotating more data to gain a more generalized model can solve this problem, but labeling location annotation for SAR images has a great cost. Domain adaptation, which is a branch of transfer learning, shares great ideas to alleviate such problem by transferring information to the unlabeled target domain (sensor) with aid of the labeled source domain (sensor). In the field of computer vision, a few people have conducted research on domain adaptation for target detection. Inoue et al. [2] uses image translation and pseudo labels to build a two-stage weakly-supervised method. Chen et al. [3] and Saito et al. [4] proposed a domain adaptive Faster R-CNN model to learn domain-invariant features. Chen et al. [5] further improve the performance with an importance-weighted GRL. Kim et al. [6] generates multiple domain to retain color information and using adversarial learning to align multiple domains. These methods mostly focus on aligning the distribution of the source and target domains and indirectly learning the characteristics of the target domain, which loses some discriminative features. What's more, when above methods are applied to the SAR field, the following two issues need to be considered: (1) Since the features of SAR images are mainly decided by edge and texture information, the methods involving color constraint cannot applied to SAR images directly. (2) Compared with optical data, few labeled training data is an important problem in SAR target detection tasks.

We proposed a cross sensor transfer learning for unsupervised SAR target detection in this paper, where data of the source sensor is labeled while data of the target sensor is unlabeled. Our method includes three stages: pixel domain diverse (PDD), multilevel feature domain alignment (MFA) and iterative self-training (IST). To add the number of labeled trainable samples, the PDD stage uses CycleGAN [7] to generate transition domain to enhance the generalization of the model from different sensors. MFA can learn the domain invariant features in the multi-layer feature space by using the idea of adversarial learning. PDD and MFA can reduce the difference at the pixel level and feature level between the data of the source sensor and the data of the target sensor. However, due to the data of the target sensor is unlabeled while the data of source sensor is labeled, the detector will affect by the back-propagation of network with labeled source data and meet the

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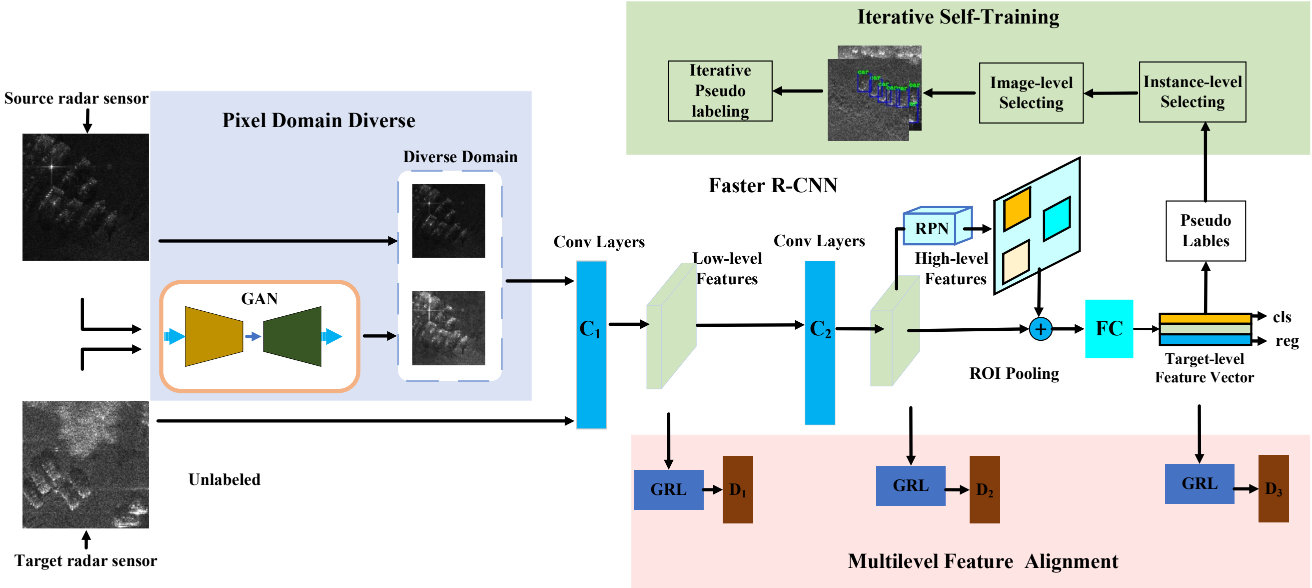


Fig. 1. The architecture of cross sensor transfer learning for unsupervised SAR target detection.

source-biased problem. To solve above issue and learn more discriminative features, we further proposed the IST that can generate and select high-quality pseudo labels directly. More specifically, the main contributions can be summarized as follows:

1) we proposed a novel cross sensor transfer learning for unsupervised SAR target detection, which can improve the performance of the unsupervised target domain with the help of supervised source domain.
2) IST, which can select pseudo-labels on target-aspect and image-aspect, is proposed to learn more discriminative features directly.
3) We introduced the PDD and MFA to reduce the discrepancy between the two domains from the pixel level and feature level.

We use two measured datasets from different radar sensors, miniSAR and FARAD. The experimental results on miniSAR and FARADSAR datasets demonstrate the effectiveness of the proposed cross sensor transfer learning for unsupervised SAR target detection based on domain adaptation.

Next, we will introduce the proposed cross sensor transfer learning for unsupervised SAR target detection in Section II, give the experimental results and analysis in Section III, and conclude this paper in Section IV.

II. PROPOSED METHOD

The details of our method will be introduced in this section. Formally, we define the data in the source sensor with labeled annotations as source domain, and the data in the target sensor without annotations as target domain. $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ represents the source domain with N_s annotated SAR images, where y_i^s is the ground-truth of x_i^s, and $\mathcal{D}_t = \{(x_j^t)\}_{j=1}^{N_t}$ represents the target domain with N_t unlabeled SAR images. The gole of our approach is to improve the performance of the unsupervised target sensor with the aid of the supervised source sensor data.

Fig.1 shows the architecture of the proposed cross sensor transfer learning based on domain adaptation. Our method includes three stages: pixel domain diverse (PDD), multilevel feature alignment (MFA) and iterative self-training (IST). We use the Faster R-CNN to build our target detection framework and uses VGG16 as the feature extractor. And next, we will introduce the structure of each stage detailly.

A. *Pixel domain diverse*

SAR images obtained by different radar sensors generally have differences in appearance, because the parameters, viewing angles, and shooting scenes of different radar sensors are often different. Some traditional methods adjust the brightness of the image by adjusting the gray value of the image. However, these methods usually cannot adjust all the images adaptively but require professionals to perform different processing based on experience., and cannot learn the data distributions. In order to automatically reduce the appearance difference of the two data domains from different sensors and increase the number of training samples, we use CycleGAN to generate images that is similar to the distribution of the target domain images. CycleGAN, an unsupervised image translation network, can generate images that looks like the images of the target domain without changing the location information of targets, which allows these data to obtain the same label as the source domain data. We call the generated data the transformation domain. However, due to the inherent problems of GAN, some generated data are not imperfect (imperfect translation problem) [6]. To alleviate the imperfect translation problem and alleviate few labeled training SAR images problem, the source domain and the transformation domain are combined to gain more data. We called the new combined data domain as diverse domain.

B. *Multilevel feature alignment*

In a deep CNNs, high-level features (extracted by high-layers of CNNs) reflect the semantic information (parking lot, trees, cars, etc.) and low-level features (extracted by low-layers of CNNs) reflect the contain structure, texture and shape information. Unlike optical images, the features of SAR images do not contain color but are mainly determined by shape, edge and texture, etc. These characters are mainly included in the low-layer feature space. Therefore, it is necessary to align features at the low-layer space between two domains. In particular, for target detection tasks, target information is also important. In Faster RCNN, the target-level information is contained in the ROI-based feature vectors. According to the above analysis, we proposed MFA to align features at multilayer space of CNNs using the thoughts of adversarial learning.

MFA aims to align features at the low-layer feature space to reduce the differences of structure, texture and shape, align features at the high-layer feature space to align the semantic distributions, and align features at the target-layer to align target distributions. We use the domain classifier and gradient reversal layer (GRL) to implement above processes. The feature extractor(VGG16) extracts low-layer features by the low-level (7th) convolution layer of VGG16 and high-layer features by the high-level (13th) convolution layer of VGG16 (C_1 and C_2 in Fig. 1, respectively). The goal of domain classifier is to distinguish the domain categories of features and the goal of feature extractor is to fool the classifier so that it cannot distinguish the domain category of the features. We use the GRL to achieve the above adversarial process, and the adversarial loss for low-layer and high-layer is as follows:

$$\mathcal{L}_{lh} = -\sum_{i,k}[d_i \log p_i^k + (1-d_i)\log(1-p_i^k)]. \quad (1)$$

Where p_i^k is the image x_i probability output of the domain classifier D_k, d_i={0,1} is the domain label. The domain classifier optimizes the parameters to minimize the loss and the feature extractor optimizes the parameters to maximize loss.

The target-level features are described as the proposal vectors (FC in Fig. 1). The adversarial process is similar to the low-level and high-level feature alignment. The target-level loss can be written as:

$$\mathcal{L}_{tar} = -\sum_{i,j}[d_i \log p_{i,j} + (1-d_i)\log(1-p_{i,j})]. \quad (2)$$

Where $p_{i,j}$ represents the probability of jth proposal of image x_i belong to diverse domain, which is the result of domain classifier D_3.

$\mathcal{L}_{regression}$ and $\mathcal{L}_{classify}$ are the Faster RCNN loss that are used to classification and regression. The final training loss of the MFA is as follows:

$$\mathcal{L} = \mathcal{L}_{regession} + \mathcal{L}_{classify} + \lambda(\mathcal{L}_{lh} + \mathcal{L}_{tar}). \quad (3)$$

Where λ is the parameter to adjust the weight of the Faster RCNN loss and the alignment loss.

C. Iterative self-training

In the previous stage, since only the diverse domain data has labels, the detector trained by back-propagation is guided by samples from diverse domain, which easily leads to domain-biased problem. Motivated by self-training method, we use pseudo labels to solve above issue. An iterative self-training method that can select pseudo-labels not only on target-aspect but also on image-aspect is proposed. Traditional self-training strategy is setting a confident threshold to select high confident proposal bounding boxes. We called this strategy as target-level strategy, since it only focuses on target bounding boxes and not considers the overall quality of pseudo labels in an image.

$$y_j = \begin{cases} 1, & p_j \ge conf. \\ 0, & Otherwise. \end{cases} \quad (4)$$

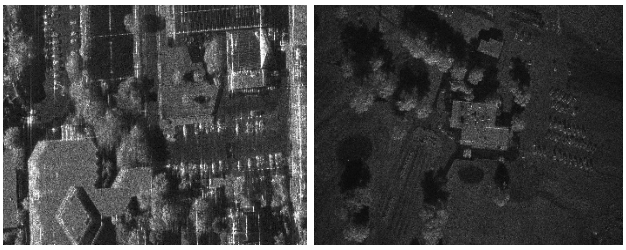
For some images, lots of false alarms will be produced when using a low confidence threshold, and vice versa. Such images with pseudo labels will harm on the network training. To reduce the negative effect, we propose an image-level selection method that can select high-quality pseudo-labels on image-aspect. Firstly, we first take the average $score_i$ of the target-level pseudo-label scores in each image as the score of the image. Then we descending sort by $score_i$ and select the top k images participating in the first iterative training.

$$score_i = \frac{1}{n_i}\sum_{j=1}^{n_i} p_{i,j}. \quad (5)$$

After the first iterative training, we annotated pseudo labels for the remaining images and we combine with the images selected in the first iteration to train the detector.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Dataset Introduction



(a) (b)

Fig. 2. Two images of FARADSAR dataset (left) and miniSAR dataset (right).

TABLE I. THE DETAILs INFORMATION OF THE FARADSAR AND THE MINISAR DATA

datasets	FARADSAR	miniSAR
Radar	FARAD	miniSAR
Mode	Stripmap	Spotlight
Band	Ka	Ku
Image size	1300×580 - 1700×1850	1638×2510
Train/test	78 for train and 28 for test	7 for train and 2 for test
Grazing Angle	29º-34º	26º-29º
Location and Scenes	Dense buildings in the University of New Mexico	Golf Course, etc. in Kirtland Air Force Base.
Year	2015	2005

We use the miniSAR and FARADSAR datasets to validate the proposed method. We use the train-set of miniSAR as source domain, use the train-set of miniSAR as target domain and use the test-set of FARADSAR to verify the performance of our method. Fig. 4 and TABLE I shows two example images and acquisition information of miniSAR FARADSAR and datasets. In our experiment, $\lambda = 0.5$ is set in Eq. (3). and the confident threshold is set to 0.5. In the IST, we set k as 50% of all images.

TABLE II. QUANTITATIVE RESULTS OF DIFFERENT UNSUPERVISED METHODS

	Precision	Recall	F1-score
Gaussian-CFAR[8]	0.2813	0.4671	0.3512
HTCN[5]	0.5182	0.8365	0.6400
DAF[3]	0.4823	0.8640	0.6191
Proposed method	0.7660	0.8104	**0.7875**

B. Comparison With Other detection Methods

To demonstrate the great performance of the our method, we compare it with unsupervised Gaussian-CFAR and other famous unsupervised domain adaptation methods, including Gaussian-CFAR, DAF and HTCN. Fig.6 gives the detection results on two test SAR images on FARADSAR respectively.

As shown in Fig.6, our method has the fewest false alarms and missing alarms compared with other methods. Moreover, for quantitative analysis, we also evaluate the overall detection results in terms of precision , recall, and F1-score in Table II. We can conclude that our method has the highest F1-score on the test SAR images, which proves that the proposed method is significantly outperforms the other above-mentioned target detection methods.

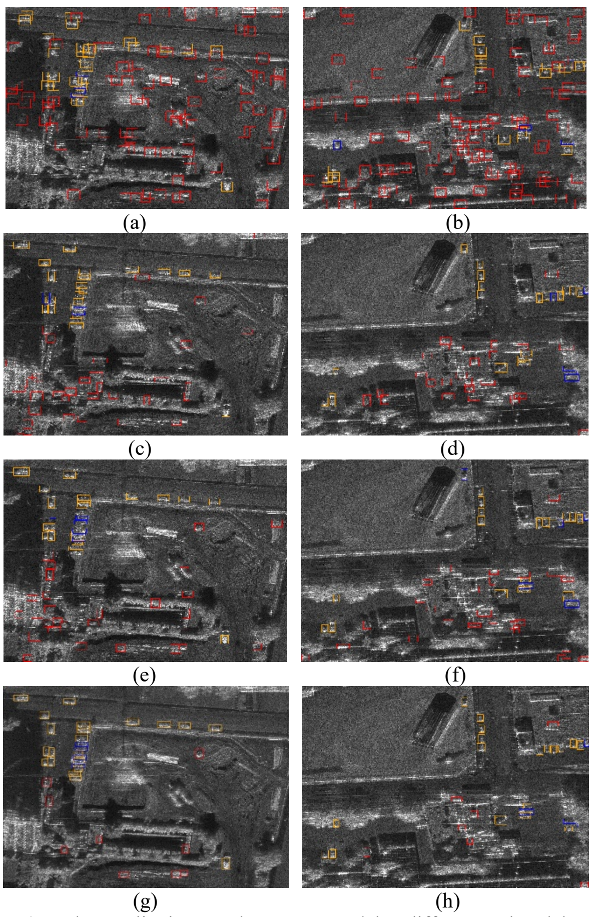


Fig. 3. The qualitative results represented by different colored boxes comparing of unsupervised methods on two test images of FARADSAR, where red rectangles denotes that the detected targets are false alarms, yellow rectangles are the result of correct detection, and blue rectangles are the missed alarms. (a) and (b) Gaussian-CFAR method. (c) and (d) HTCN method. (e) and (f) DAF method. (g) and (h) Our method.

C. Ablation Study

We design the ablation studies to verify three stages of our cross sensor transfer learning method based on domain adaptation. We give quantitative results of the impact of each stage on the detection performance in TABLE III, where PDD, MFA, and IST represent the name of each stage and '+' indicates which stage or the stage combination to use.

As shown in Table III, three stages can all improve the performance of target detection. From the table III we can see that each stage can reduce missed alarms. According to the ablation study of the IST stage, we can conclude that the better the detector performance of previous stages, the better the quality of the pseudo labeles. According to the whole result of TABLE III, we can find that our approach can achieve the best performance of target detection by combining the PDD, MFA, and IST at the same time.

TABLE III. QUANTITAVE RESULTS OF ABLATION STUDIES

	PDD	MFA	IST	Precision	Recall	F1-score
No-DA				0.4491	0.8133	0.5787
Ours	+			0.5567	0.8553	0.6744
	+		+	0.6004	0.8538	0.7050
		+	+	0.6496	0.7750	0.7067
	+	+		0.6912	0.7328	0.7108
	+	+	+	0.7660	0.8104	**0.7875**

IV. CONCLUSION

In this paper, for inconsistent data distribution across sensors we proposed a cross sensor transfer learning for unsupervised SAR target detection, where data of the source sensor is labeled while data of the target sensor is unlabeled. With our approach, the SAR target detection performance of the unsupervised target domain is improved significantly. The proposed method constructs a three-stage transfer learning framework based on domain adaptation for unsupervised SAR target detection. PDD and MFA can alleviate the performance drop by align distributions at pixel and feature level. PDD can generate transition domain through GANs to add the number of labeled trainable samples and enhance the generalization of the model from different sensors from different sensors. MFA can learn the domain invariant features in the multi-layer feature space by using the idea of adversarial learning. However, due to the data of the target sensor is unlabeled while the data of source sensor is labeled, the detector will affect by the back-propagation of network with labeled source data. Motivated by self-training method, we use pseudo labels to solve above issue. An iterative self-training method that can select pseudo-labels not only on target-aspect but also on image-aspect is proposed. The experimental results shows that the proposed method is significantly outperforms the other above-mentioned target detection methods, which demonstrates the effectiveness of our cross sensor transfer learning for unsupervised SAR target detection method.

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